

PHISHING ATTACK SIMULATION AND DETECTION SYSTEM

Name : Hemamalini Githolla

Date : 19/03/2026

Domain : Cybersecurity

CONTENTS

- 1. Executive Summary**
- 2. Objective**
- 3. Tools & Technologies**
- 4. Methodology**
- 5. Email Analysis using Security Tools**
- 6. Python- Based Phishing Detection System**
- 7. Results and Comparative Analysis**
- 8. Key Findings**
- 9. Conclusion**
- 10. Future Enhancements**

Executive Summary

Phishing attacks remain one of the most effective social engineering techniques used by attackers to compromise user credentials and sensitive information. This project demonstrates both the **simulation of a phishing attack** and the **development of an intelligent detection system** to identify such threats.

A phishing email was designed to mimic a legitimate banking alert, incorporating common attack characteristics such as urgency, deceptive links, and generic messaging. In parallel, a legitimate email was analyzed to establish a baseline for comparison.

A Python-based detection engine was developed to analyze emails across multiple dimensions, including **header authentication (SPF, DKIM, DMARC), content patterns, and URL intelligence**. The system uses a **weighted risk scoring model** to classify emails into *High Risk Phishing*, *Suspicious*, or *Legitimate*.

To validate findings, industry-recognized tools such as Google Admin Toolbox, MXToolbox, and VirusTotal were utilized.

Objective

- Simulate a realistic phishing attack scenario
- Analyze email headers and authentication mechanisms
- Identify phishing indicators in content and URLs
- Develop a rule-based phishing detection system
- Compare phishing and legitimate email characteristics
- Demonstrate practical cybersecurity analysis skills

Tools & Technologies Used

- **Programming Language:** Python
- **Libraries:** `email`, `re`, `urllib`, `json`
- **Analysis Tools:**
 - Google Admin Toolbox
 - MXToolbox
 - VirusTotal

Methodology

Phishing Attack Simulation

A phishing email was crafted to replicate a financial security alert. The email included:

- A subject line indicating urgency (e.g., “Suspicious Transaction Detected”)
- A generic greeting to avoid personalization
- A malicious verification link
- Language designed to induce panic and immediate action

This simulated real-world phishing tactics used by attackers.

Legitimate Email Collection

A real email header was extracted from a trusted source (e.g., Gmail). This email contained:

- Valid SPF, DKIM, and DMARC authentication
- Proper domain alignment
- Normal, non-threatening content

This served as a baseline for comparison.

Email Analysis Using Security Tools

Phishing Email Analysis

Google Admin Toolbox Messageheader Help						
MessageId	20260210102015.12345@mail.fakebank.com					
Created at:	2/10/2026, 10:20:15 AM GMT+5:30 (Delivered after 5 mins)					
From:	"Bank Security Team" <fraud-team@secure-verification-bank.com>					
To:	victim.user@gmail.com					
Subject:	Urgent: Suspicious Transaction Detected					
SPF:	fail with IP Unknown! Learn more					
DKIM:	fail with domain Unknown! Learn more					
DMARC:	fail Learn more					
#	Delay	From *		To *	Protocol	Time received
0	5 mins	mail.fakebank.com	→	 mx.google.com	ESMTPS	2/10/2026, 10:25:31 AM GMT+5:30
ANALYZE ANOTHER HEADER						

Figure 1: Phishing email header analysis showing SPF, DKIM, and DMARC authentication failures.

Explanation:

The phishing email failed all major authentication checks, indicating that the sender domain was not authorized. This is a strong indicator of spoofing.

New TabGoogle DocsPhishing Em...Phishing Em...Project Sele...hacker ai - G...Messagehe...TASK 2Email Head...

mxtoolbox.com/Public/Tools/Email-headers.aspx?uid=1fa8e75-5876-4cfb-a7ab-9ee36816715c

MX TOOLBOX[®]

SUPERTOOL

PricingToolsDelivery CenterMonitoringProductsBlogSupportLogin

SuperToolMX LookupBlacklistsDMARCDiagnosticsEmail HealthDNS LookupAnalyze HeadersAll Tools

Header Analyzed

Email Subject: Urgent: Suspicious Transaction Detected

Analyze New Header

Copy/Paste Warning

Copy/Pasting a header works for most people, but sometimes it can cause problems with things like DKIM Validation. For the best results, use our [Email Deliverability tool](#)

Delivery Information

DMARC Compliant (No DMARC Record Found)

SPF Alignment

SPF Authenticated

DKIM Alignment

DKIM Authenticated

Relay Information

Received

0 seconds

Delay:

Your IP is 223.169.77.85 | Contact Terms & Conditions Site Map Security API Privacy Phone: (866) 696-6952 | © Copyright 2004-2026, MxToolBox, Inc. All rights reserved. US Patents 19039353 B2 & 11461738 B2

Type here to search32°C SunnyENG IN12:28 PM3/19/2026

New TabGoogle DocsPhishing Em...Phishing Em...Project Sele...hacker ai - G...Messagehe...TASK 2Email Head...

mxtoolbox.com/Public/Tools/Email-headers.aspx?uid=1fa8e75-5876-4cfb-a7ab-9ee36816715c

DKIM Authenticated

Relay Information

Received

0 seconds

Delay:

From mail.fakebank.com to mx.google.com

00.20.40.60.801.01.2

Relay (Seconds)

Hop	Delay	From	By	With	Time (UTC)	Blacklist
1	*	mail.fakebank.com 185.234.217.89	mx.google.com	ESMTPS	[Mon, 10 Feb 2026 10:25:31 +0530 (IST)]	

Gmail & Yahoo are now requiring DMARC - Get yours setup with Delivery Center

SPF and DKIM Information

dmARC:secure-verification-bank.com

Hide

Solve Email Delivery Problems

Test

Result

DMARC Record PublishedNo DMARC Record found

Your IP is 223.169.77.85 | Contact Terms & Conditions Site Map Security API Privacy Phone: (866) 696-6952 | © Copyright 2004-2026, MxToolBox, Inc. All rights reserved. US Patents 19039353 B2 & 11461738 B2

Type here to search32°C SunnyENG IN12:28 PM3/19/2026

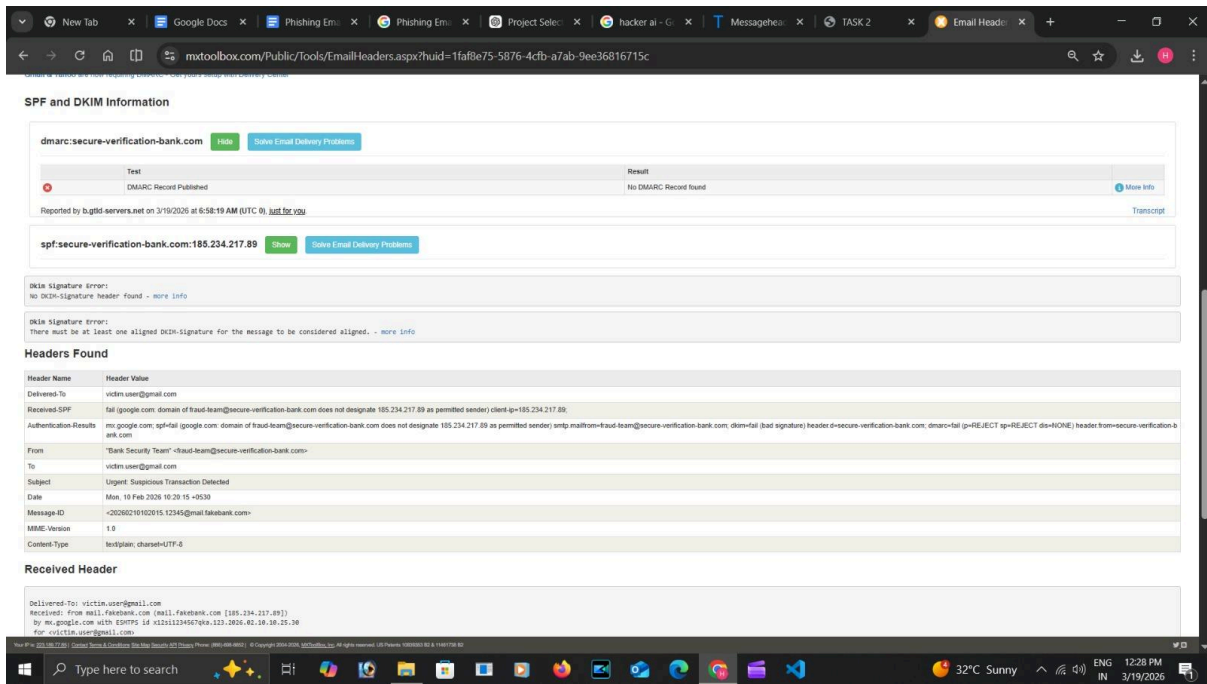


Figure 2: MXToolbox analysis highlighting suspicious sending server and blacklist status.

The originating server was identified as suspicious and not associated with a legitimate organization.

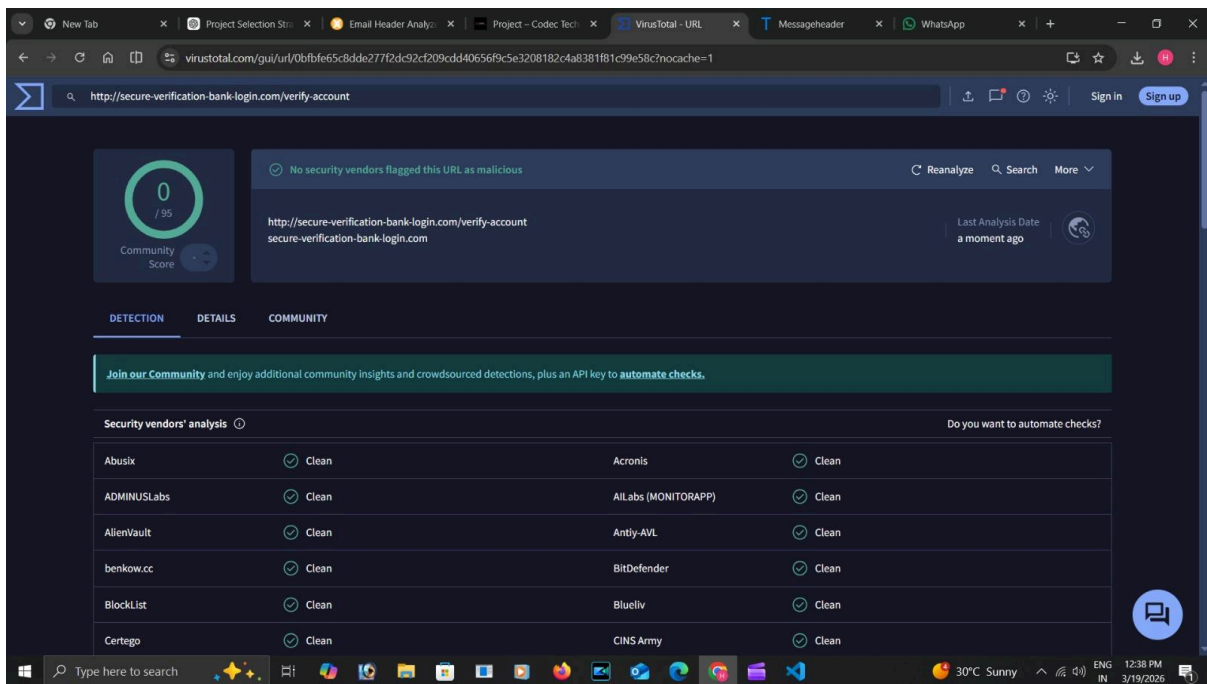


Figure 3: URL scan result showing no immediate detection but potential risk.

Explanation:

Although the URL was not flagged, newly created phishing domains often evade detection initially.

Legitimate Email Analysis

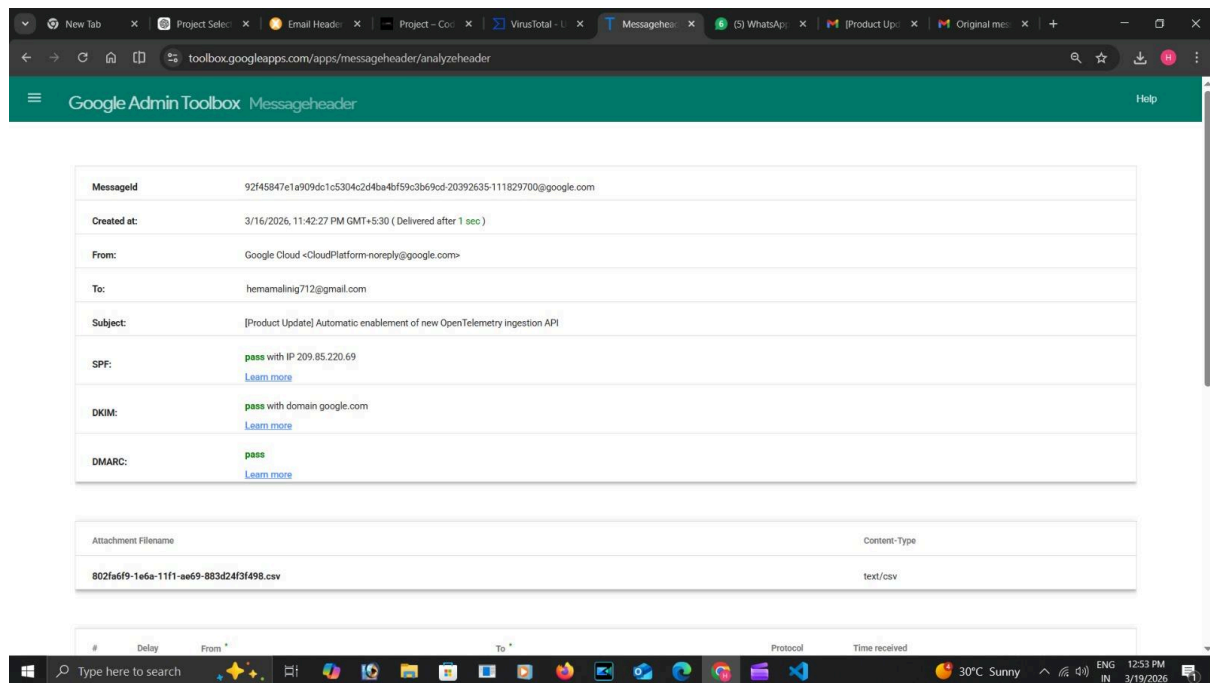


Figure 4: Legitimate email showing successful SPF, DKIM, and DMARC validation.

Explanation:

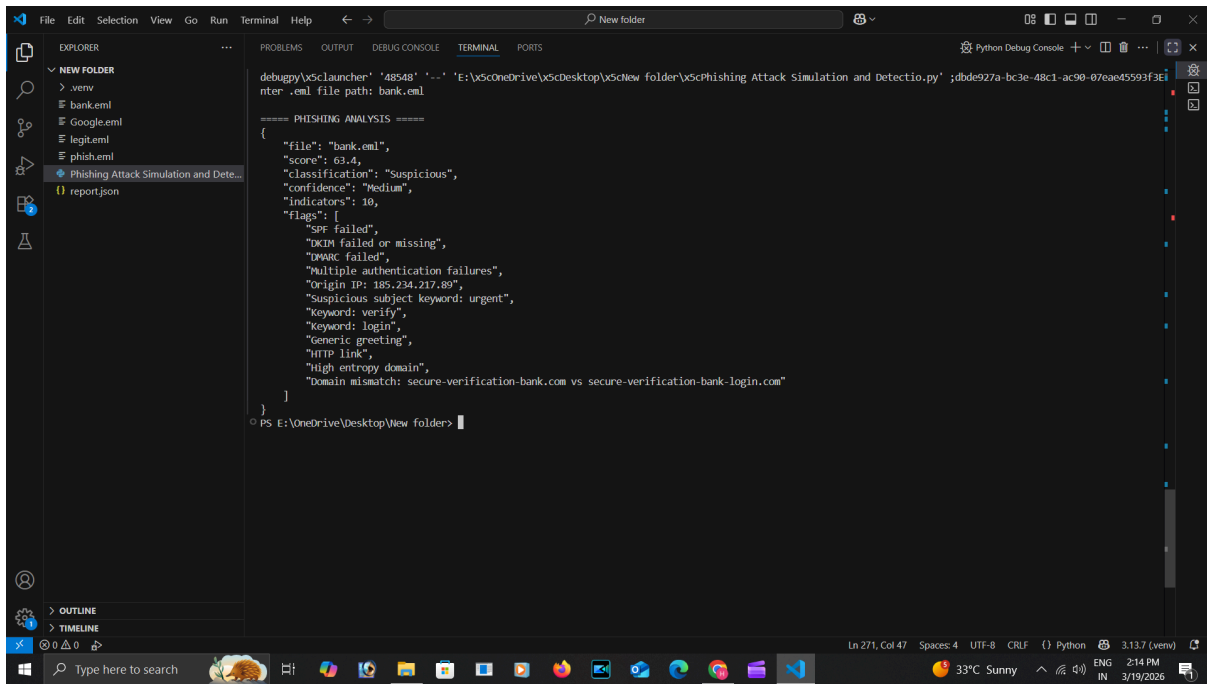
All authentication checks passed, confirming the sender's legitimacy and domain integrity.

Python-Based Phishing Detection System

A custom phishing detection system was developed using Python to automate email analysis.

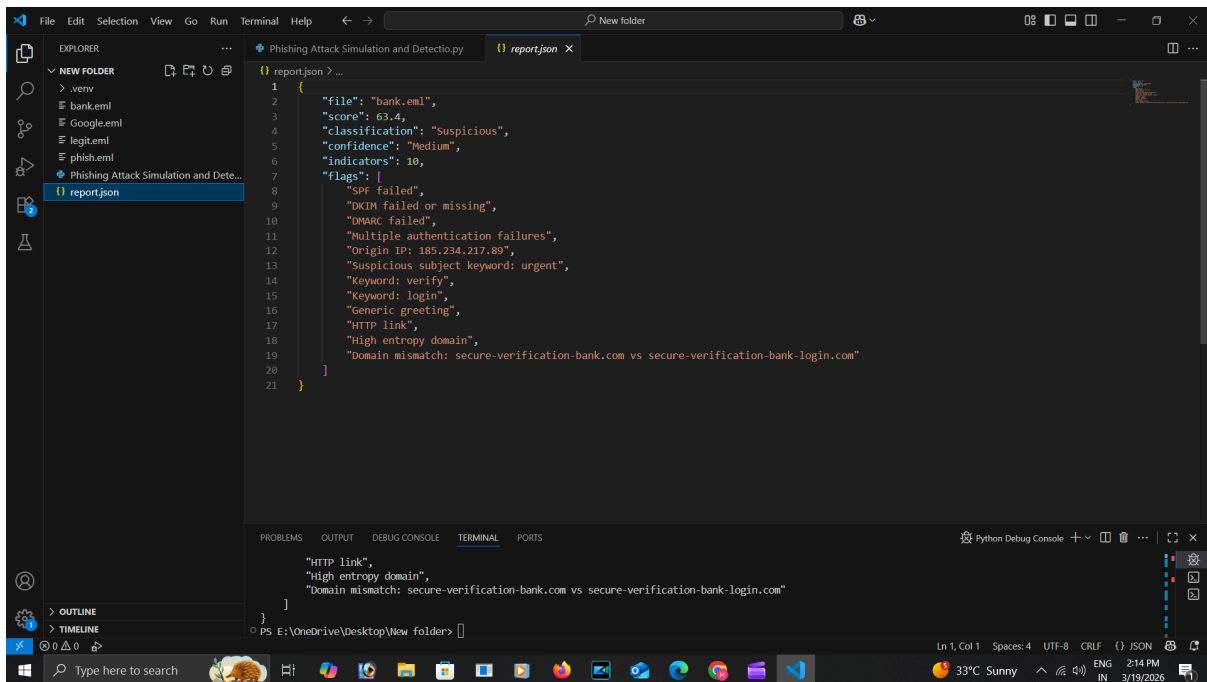
Key Features:

- **Header Analysis:** Detects SPF, DKIM, and DMARC failures
- **Content Analysis:** Identifies phishing keywords, urgency, and generic greetings
- **URL Analysis:** Evaluates domains for suspicious patterns, entropy, and mismatches
- **Risk Scoring Model:** Combines multiple indicators into a final score
- **Classification Engine:** Categorizes emails into risk levels



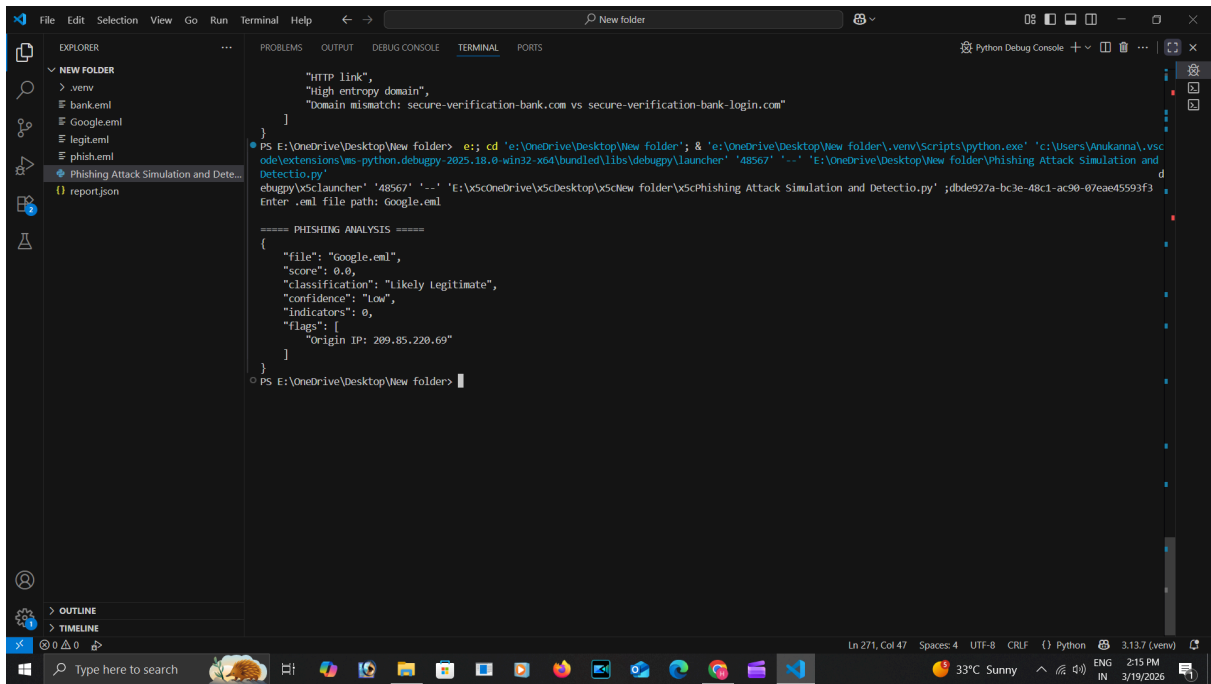
```
debugpy\vsclauncher' '48548' '-' 'E:\OneDrive\Desktop\New folder\vscodePhishing Attack Simulation and Detectio.py' ;dbde927a-bc3e-48c1-ac90-07eae45993f3E
nter .eml file path: bank.eml

==== PHISHING ANALYSIS ====
{
  "file": "bank.eml",
  "score": 63.4,
  "classification": "Suspicious",
  "confidence": "Medium",
  "indicators": 10,
  "flags": [
    "SPF failed",
    "DKIM failed or missing",
    "DMARC failed",
    "Multiple authentication failures",
    "Origin IP: 185.234.217.89",
    "Suspicious subject keyword: urgent",
    "Keyword: verify",
    "Keyword: login",
    "Generic greeting",
    "HTTP link",
    "High entropy domain",
    "Domain mismatch: secure-verification-bank.com vs secure-verification-bank-login.com"
  ]
}
PS E:\OneDrive\Desktop\New folder>
```



```
1 {
2   "file": "bank.eml",
3   "score": 63.4,
4   "classification": "Suspicious",
5   "confidence": "Medium",
6   "indicators": 10,
7   "flags": [
8     "SPF failed",
9     "DKIM failed or missing",
10    "DMARC failed",
11    "Multiple authentication failures",
12    "Origin IP: 185.234.217.89",
13    "Suspicious subject keyword: urgent",
14    "Keyword: verify",
15    "Keyword: login",
16    "Generic greeting",
17    "HTTP link",
18    "High entropy domain",
19    "Domain mismatch: secure-verification-bank.com vs secure-verification-bank-login.com"
20  ]
21 }
```

Figure 5: Detection engine output classifying the phishing email as High Risk.



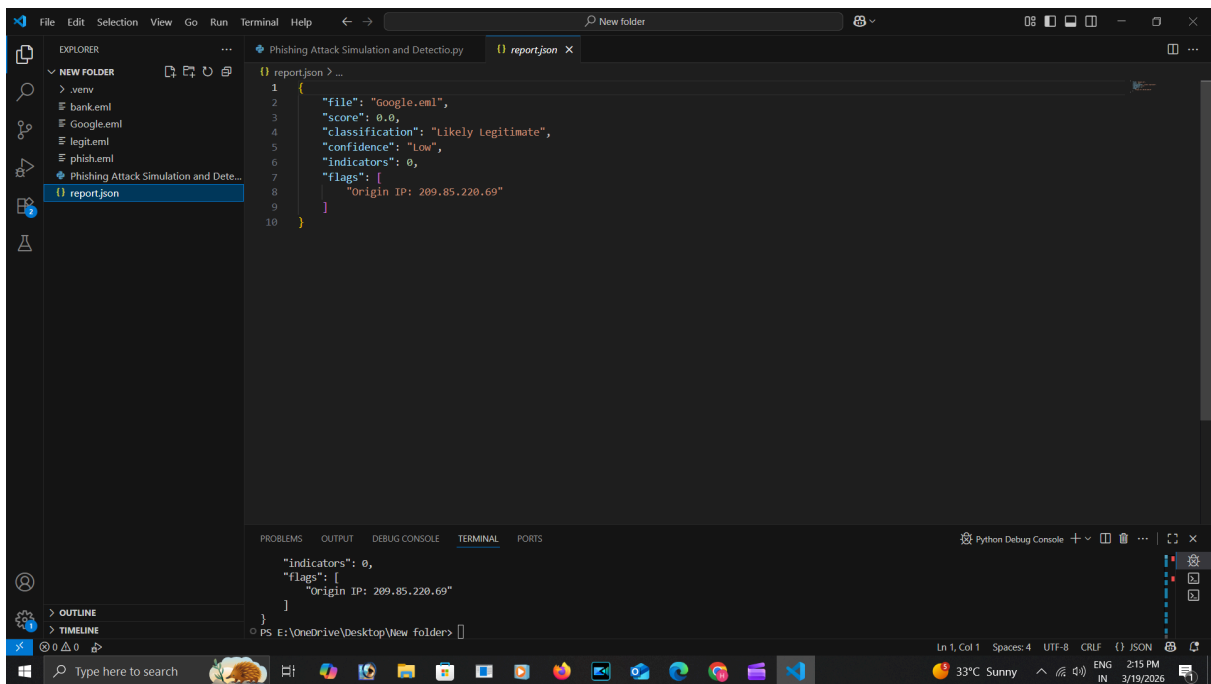
```
File Edit Selection View Go Run Terminal Help
New folder

EXPLORER
NEW FOLDER
> .venv
  bank.eml
  Google.eml
  legit.eml
  phish.eml
  Phishing Attack Simulation and Detection.py
  report.json

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Python Debug Console

PS E:\OneDrive\Desktop\New folder> e; cd 'e:\OneDrive\Desktop\New folder'; & 'e:\OneDrive\Desktop\New folder\.venv\Scripts\python.exe' 'c:\Users\Anukamal\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher' '48567' '-.' 'E:\OneDrive\Desktop\New folder\Phishing Attack Simulation and Detection.py'
Enter .eml file path: Google.eml

==== PHISHING ANALYSIS ====
{
  "file": "Google.eml",
  "score": 0.0,
  "classification": "Likely legitimate",
  "confidence": "Low",
  "indicators": 0,
  "flags": [
    "Origin IP: 209.85.228.69"
  ]
}
PS E:\OneDrive\Desktop\New folder>
```



```
File Edit Selection View Go Run Terminal Help
New folder

EXPLORER
NEW FOLDER
> .venv
  bank.eml
  Google.eml
  legit.eml
  phish.eml
  Phishing Attack Simulation and Detection.py
  report.json

report.json x
1 {
2   "file": "Google.eml",
3   "score": 0.0,
4   "classification": "Likely legitimate",
5   "confidence": "Low",
6   "indicators": 0,
7   "flags": [
8     "Origin IP: 209.85.228.69"
9   ]
10 }

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Python Debug Console

"indicators": 0,
"flags": [
  "Origin IP: 209.85.228.69"
]
}
PS E:\OneDrive\Desktop\New folder>
```

Figure 6: Detection engine output classifying the legitimate email as safe.

7. Results and Comparative Analysis

Comparison Table

Feature	Phishing Email	Legitimate Email
SPF	Failed	Passed
DKIM	Failed	Passed
DMARC	Failed	Passed
Domain Authenticity	Suspicious	Trusted
Content Style	Urgent / Threatening	Normal
URL Behavior	Suspicious	Safe
Detection Result	High Risk	Legitimate

Key Findings

- Authentication failures are strong indicators of phishing
- Phishing emails rely heavily on urgency and emotional manipulation
- Suspicious domains and misleading URLs are common attack vectors
- Combining multiple detection layers significantly improves accuracy

Conclusion

This project successfully demonstrates both the **creation of a phishing attack scenario** and the **development of an effective detection mechanism**.

The multi-layer detection system proved capable of identifying phishing characteristics using a combination of header validation, content inspection, and URL intelligence. This approach reflects real-world cybersecurity practices and highlights the importance of layered defense strategies.

Future Enhancements

- Integration with real-time threat intelligence services
- Machine learning-based phishing classification
- Deployment as an email filtering system
- Browser extension for phishing prevention